Ogbonna Arinze

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SUMMARY

Lead Fullstack Engineer with 8 years of professional experience architecting and delivering modern web applications across healthcare, fintech, civic tech, and enterprise environments. Deep expertise in **React**, Next.js, and TypeScript on the frontend, with proven backend capabilities in **Node.js REST APIs**, NestJS, and PostgreSQL. Experienced technical leader who defines software architecture, drives technology selection, establishes code quality standards, and mentors engineering teams. Proficient in leveraging **modern AI tools** for accelerated development, with hands-on experience building production RAG/LLM systems. Brings direct healthcare technology experience from Genentech (Roche), building clinical data products that bridge complex backend systems with intuitive user interfaces.

CORE COMPETENCIES

- **Fullstack Architecture:** Software architecture design, scalable component systems, design systems, monorepos (Nx, pnpm workspaces), SSR/SSG, and design pattern proficiency.
- **Frontend:** React (in-depth), Next.js, Vue.js, TypeScript, Redux, Zustand, Tailwind CSS, Radix UI, Storybook, and Webpack/Vite/Rollup.
- **Backend & APIs:** Node.js REST API development, NestJS, PostgreSQL, MongoDB, Prisma, GraphQL, JWT authentication, and role-based access control.
- **AI & Development Tools:** AI-powered development assistants (Claude, GitHub Copilot), RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), and agentic workflows.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, ESLint, Prettier, CI-driven QA, and code quality standards enforcement.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, Docker Compose, GitHub Actions, CI/CD pipelines, and Nx monorepo orchestration.
- **Data Visualization:** Recharts, react-chartjs, dynamic dashboards, and interactive chart systems for complex data presentation.
- **Leadership:** Technical leadership, architecture reviews, cross-functional collaboration, mentorship, Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension, defining architecture decisions, technology selection, and implementation strategy for the healthcare platform.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a conversational interface.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams in the healthcare organization.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with TypeScript and React Query, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing **lazy loading, code splitting**, dynamic imports, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and **code quality standards**.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with software architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.

- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and higher release confidence.
- Established engineering standards for **code quality**, testing, and deployment using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices.
- Mentored junior engineers and guided frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing conventional socket.io methods for faster data updates and a more responsive interface.
- Built dynamic **data visualization dashboards** using React and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **Node.js RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, collaborating closely with the business team to ensure technical solutions met product needs.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.
- Oversaw the professional development of two interns, guiding them through frontend and backend projects with technical mentorship and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **Node.js**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **Node.js** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered platform for Nigerian budget analysis and corruption tracking. Multi-agent LLM pipeline analyzes 700+ budget documents (708K+ vector embeddings) across 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with 22 interactive chart types (Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with pnpm workspaces.
- Designed and implemented a **multi-agent AI system** using the Mastra framework with 5 specialized agents and a three-stage intent classification pipeline, powered by a **hybrid RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, Reciprocal Rank Fusion, and **Cohere** reranking.
- Built the **Node.js/NestJS** backend with **PostgreSQL 16**, **Prisma v7**, SSE streaming with 15s keepalive heartbeats, phone OTP authentication, and **REST API** endpoints for chat, user management, and analytics.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and Traefik reverse proxy routing.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, dark mode, URL state management, responsive mobile layouts, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*